

UNIFORM SHEAFY TATE RINGS THAT ARE NOT STABLY UNIFORM

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ABSTRACT. Buzzard–Verberkmoes and Mihara show that a stably uniform Tate Huber ring is sheafy. We construct examples of uniform and strongly sheafy Tate rings which are not stably uniform, answering Question 7 of Kedlaya’s *Nonarchimedean Scottish Book*. In the first example a rational localisation is non-reduced, whilst in the second the rational localisation is an integral domain but is not uniform. The two main results are due to ChatGPT 5.6 Sol. Both constructions and the rational localisations witnessing failure of stable uniformity have been formalised in Lean 4; the strong-sheafiness formalisation is complete for the second example and for the Laurent-series case of the first.

1. INTRODUCTION

The examples of Buzzard–Verberkmoes and Mihara show that uniformity can be lost on a rational chart and, separately, that a uniform Tate Huber ring need not be sheafy [4, 18]. They leave open whether every uniform sheafy Tate Huber ring is stably uniform. Hansen and Kedlaya ask this explicitly [8, Remark 3.16]; it is also Question 7 of Kedlaya’s *Nonarchimedean Scottish Book* [19]. We answer this question in the negative.

Following Hansen–Kedlaya [8, Definition 4.1], a Huber ring R is *strongly sheafy* if $R\langle V_1, \dots, V_s \rangle$ is sheafy for every $s \geq 0$. In fact, both examples below are strongly sheafy.

The examples and proofs we will describe here were found with the help of ChatGPT 5.6 Sol. In Section 9 we describe how we used ChatGPT in the search for these examples. The Lean formalisation was carried out by Claude Code, and appendix A records the formal definitions and theorem statements. This article was prepared with the help of AI from an initial human-written skeleton.

Our first example is what the AI calls the “finite-jet pullback” example. This is a non-noetherian uniform integral domain for which a certain rational localisation is non-reduced, and hence is no longer uniform. Let k be a complete discretely valued nonarchimedean field, with valuation ring k° , uniformizer ϖ , and residue field $\tilde{k}$. Put

$$L_0 = k^\circ \langle W, W^{-1} \rangle, \quad B_0 = k^\circ \langle W, Q \rangle / (Q^2),$$

$$C_0 = L_0 \langle Q \rangle, \quad D_0 = L_0 \langle Q \rangle / (Q^2),$$

where $k^\circ \langle W, W^{-1} \rangle$ means $k^\circ \langle W, V \rangle / (WV - 1)$. There are natural maps $B_0 \rightarrow D_0$ and $C_0 \twoheadrightarrow D_0$. Define

$$A_0 = B_0 \times_{D_0} C_0, \quad A = A_0[1/\varpi],$$

and endow A with the topology defined by the ring of definition A_0 .

The main result is the following.

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Theorem 1.1. *The ring A is a complete uniform non-noetherian Tate k -algebra and an integral domain, with $A^\circ = A_0$. It is strongly sheafy; in particular, the Huber pair (A, A°) is sheafy. However,*

$$A \left\langle \frac{W}{\varpi} \right\rangle \cong k\langle X, Q \rangle / (Q^2), \quad X = \frac{W}{\varpi},$$

so A is not stably uniform.

The key element in this example is that after the rational localisation $A \rightarrow A\langle W/\varpi \rangle$, Q^2 vanishes whereas Q does not.

This construction also gives a new example solving Problems 24 and 28 of the *Nonarchimedean Scottish Book*. We recall the two problems and give the argument in section 7. These results have also been formalised in Lean.

The proof uses Milnor squares of complete Tate rings. Recall that a cartesian square of rings with a surjective leg is called a Milnor square [17, Section 2]; the complete Tate setting is used by Kerz–Saito–Tamme in their work on analytic K -theory [14, Section 3].

Definition 1.2. A commutative square of complete Tate k -algebras

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

is a *strict Milnor square* if it is cartesian, the sequence of underlying topological k -vector spaces

$$0 \longrightarrow A \longrightarrow B \oplus C \xrightarrow{(b,c) \mapsto \bar{b} - \bar{c}} D \longrightarrow 0$$

is strict exact, and $C \rightarrow D$ is a strict surjection.

The maps $C \rightarrow D$ and $B \oplus C \rightarrow D$ are continuous surjections between complete normed k -vector spaces, and A has the subspace topology as the kernel of the second. The nonarchimedean open mapping theorem [14, Lemma 3.1] therefore shows that the square defining A is strict. Our main input is that, for this finite-jet square, the strict exact row persists after completed rational localisation and is compatible with refinement. This transfers sheafiness from B, C, D to A , giving new ways for proving that these rings are sheafy. The second example shows that the failure of stable uniformity need not come from a non-reduced localisation.

Remark 1.3 (Relation with earlier work). It is difficult to understand exactly what sources an AI agent such as ChatGPT may have used in doing this work. In the interest of giving appropriate credit, we have listed here closely related work that may have been used in the development of the results here.

Buzzard–Verberkmoes construct a uniform affinoid ring with a rational localisation which is not uniform and, separately, a uniform affinoid ring whose structure presheaf is not a sheaf [4, Propositions 17 and 18]. Mihara gives further examples of both phenomena [18]. These examples do not give a uniform sheafy ring which is not stably uniform.

Ben-Bassat–Kretnizer prove strict two-term resolutions for Weierstrass and Laurent localisations, and reduce a general rational localisation to these cases [2, Lemmas 5.13–5.14]. Bambozzi–Kretnizer formulate rational localisation in terms of Koszul complexes and prove strict Koszul regularity [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. The bounded-denominator estimates in Lemma 5.1 are the form needed to show that rational localisation preserves the finite-jet Milnor square. The resulting uniform sheafy domain which is not stably uniform appears to be new.

The weighted-parity construction of section 8 is closer to the infinite monomial-support examples of Buzzard–Verberkmoes and Mihara. Its additional feature is that rational localisation can be calculated in a noetherian finite-variable subring.

2. CONVENTIONS AND NOTATION

We follow Huber for Huber pairs, adic spectra, rational subsets, and rational localisation [11], and use Wedhorn’s account for strong noetherianity and sheafiness [21, Proposition and Definition 6.36, Section 8.1]; the bounded-denominator language for strictness follows Buzzard–Verberkmoes [4], and the quotient presentation of a rational localisation is also discussed by Hansen–Kedlaya [8].

- *Tate topology.* Throughout, E denotes a complete Tate k -algebra. In the discretely valued setting of this paper, this means that E contains a topologically nilpotent unit and admits an open bounded k° -subalgebra E_0 such that

$$E = E_0[1/\varpi], \quad \{\varpi^n E_0\}_{n \geq 0}$$

is a basis of neighbourhoods of zero. Such an E_0 is called a *ring of definition*. We always choose it closed, hence ϖ -adically complete. A subset of E is bounded if it is contained in $\varpi^{-r} E_0$ for some $r \geq 0$.

- *Uniformity and noetherianity.* We write E° for the ring of power-bounded elements. The ring E is *uniform* if E° is bounded, and *strongly noetherian* if $E\langle Z_1, \dots, Z_s \rangle$ is noetherian for every $s \geq 0$. A Tate ring is *stably uniform* if every rational localisation is uniform.
- *Huber pairs and sheafiness.* A *ring of integral elements* is an open subring $E^+ \subseteq E^\circ$ which is integrally closed in E . We call the Huber pair (E, E^+) *sheafy* when its structure presheaf, valued in complete topological rings, satisfies the sheaf condition. Thus, for a finite rational cover $U = \bigcup_i U_i$, restriction identifies $\mathcal{O}(U)$ homeomorphically with the subspace of $\prod_i \mathcal{O}(U_i)$ consisting of tuples whose restrictions to every $U_i \cap U_j$ agree.
- *Strictness.* We use the following bounded-denominator form of strictness. Let $u : M \rightarrow N$ be a continuous k -linear map, and let $M_0 \subset M$ and $N_0 \subset N$ be open bounded k° -submodules. Then u is strict if and only if there exists $a \geq 0$ such that

$$\varpi^a(u(M) \cap N_0) \subset u(M_0).$$

For a surjection this becomes $\varpi^a N_0 \subset u(M_0)$. For an injection it says that the inverse image of N_0 is bounded in M . This is the criterion used by Buzzard–Verberkmoes [4, Lemma 2]. We use the term *strict Milnor square* in the sense of Definition 1.2; the underlying algebraic terminology is standard [17, 14].

- *Rational localisation.* We take every rational datum to have at least one numerator; an empty list may be padded by $f_1 = 0$. Let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum in E , so the ideal generated by $g, f_1, \dots, f_m$ is open. Since E is a Tate k -algebra, this ideal is the unit ideal. After multiplying the entire datum by a common power of ϖ , we may assume that all entries lie in E_0 .

The rational localisation E_α is the ϖ -adic completion of $E[1/g]$ for the ring of definition generated by

$$E_0 \left[\frac{f_1}{g}, \dots, \frac{f_m}{g} \right].$$

For such a ring of definition R , we use the notation

$$\widehat{R} = \varprojlim_{n \geq 1} R/\varpi^n R, \quad \ker(R \longrightarrow \widehat{R}) = \bigcap_{n \geq 1} \varpi^n R.$$

E_α is independent of the ring of definition and of the presentation of the rational domain, and it is transitive under rational refinement. Put

$$T_E = E\langle T_1, \dots, T_m \rangle = E_0\langle T_1, \dots, T_m \rangle[1/\varpi], \quad r_i = gT_i - f_i,$$

and

$$I_{E,\alpha} := (r_1, \dots, r_m) = \text{im} \left(T_E^m \xrightarrow{d_{1,E}} T_E \right) \subset T_E, \quad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i (gT_i - f_i),$$

Then

$$E_\alpha \cong T_E / \overline{I_{E,\alpha}}. \quad (1)$$

Here the bar is necessary until the ideal has been shown to be closed. This is the standard quotient presentation of rational localisation; see [11, Notations, (1.2), Proposition 1.3, Lemma 1.5] and [8, Definition 3.6].

3. THE FINITE-JET RING

In this section we identify the finite-jet algebra with a concrete subring of C and prove that it is a complete uniform domain. Recall that

$$L_0 = k^\circ\langle W, W^{-1} \rangle, \quad B_0 = k^\circ\langle W, Q \rangle / (Q^2),$$

$$C_0 = L_0\langle Q \rangle, \quad D_0 = L_0\langle Q \rangle / (Q^2),$$

where $k^\circ\langle W, W^{-1} \rangle$ denotes $k^\circ\langle W, V \rangle / (WV - 1)$. The natural maps $B_0 \rightarrow D_0$ and $C_0 \twoheadrightarrow D_0$ allow us to define

$$A_0 = B_0 \times_{D_0} C_0.$$

We write

$$A = A_0[1/\varpi], \quad L = L_0[1/\varpi], \quad B = B_0[1/\varpi], \quad C = C_0[1/\varpi], \quad D = D_0[1/\varpi].$$

Each of these Tate rings carries the topology defined by the corresponding subscript-zero ring.

The quotient map $C_0 \rightarrow D_0$ has a continuous k° -linear section

$$\sigma : D_0 \longrightarrow C_0, \quad \sigma(f_0 + Qf_1) = f_0 + Qf_1,$$

obtained by viewing the unique representative of Q -degree at most one as an element of C_0 . Consequently

$$0 \longrightarrow A_0 \longrightarrow B_0 \oplus C_0 \longrightarrow D_0 \longrightarrow 0 \quad (2)$$

is exact, and A_0 is ϖ -adically complete. After inverting ϖ we obtain a strict Milnor square

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D. \end{array}$$

The map $B_0 \rightarrow D_0$ is injective. Projection to C_0 therefore identifies A_0 with the subring

$$A_0 = \{ f_0(W) + Qf_1(W) + Q^2h(W, W^{-1}, Q) : f_0, f_1 \in k^\circ\langle W \rangle, h \in C_0 \}. \quad (3)$$

Equivalently, A_0 consists of the restricted series

$$\sum_{(a,b) \in S} c_{a,b} W^a Q^b, \quad S = \{(a,b) \in \mathbb{Z} \times \mathbb{N} : b \leq 1 \Rightarrow a \geq 0\},$$

with $c_{a,b} \in k^\circ$. Here restricted means that, for every n , only finitely many coefficients are nonzero modulo ϖ^n . The set S is closed under addition, so this is a subring of C_0 . The same description with coefficients in k gives $A \subset C$.

Equip A, B, C, D with the coefficientwise Gauss norms whose unit balls are A_0, B_0, C_0, D_0 , respectively. Write

$$\iota_B : A \rightarrow B, \quad \iota_C : A \rightarrow C, \quad \rho_B : B \rightarrow D, \quad \rho_C : C \rightarrow D$$

for the structural maps, and give A the restricted Gauss norm from C . For $a = f_0 + Qf_1 + Q^2h$, the coefficient formulas give

$$\|\iota_C(a)\|_C = \|a\|_A, \quad \|\iota_B(a)\|_B \leq \|a\|_A, \quad \|\rho_B(b)\|_D \leq \|b\|_B, \quad \|\rho_C(c)\|_D \leq \|c\|_C. \quad (4)$$

Thus every structural map is norm-nonincreasing.

Proposition 3.1. *The ring A is a complete uniform non-noetherian Tate k -algebra and an integral domain. Moreover,*

$$A^\circ = A_0.$$

Proof. The Gauss norm on $L = k\langle W, W^{-1} \rangle$ is multiplicative, and the restricted Gauss norm on $C = L\langle Q \rangle$ is therefore multiplicative as well. In particular, C is a domain, and hence so is its subring A . The constant- and linear-coefficient maps $C \rightarrow L$ are continuous, and $k\langle W \rangle$ is closed in $L = k\langle W, W^{-1} \rangle$ for the Gauss norm. Hence the condition that these two coefficients lie in $k\langle W \rangle$ defines a closed subring of C , so A is complete. The image of ϖ in A is a topologically nilpotent unit, and therefore A is a Tate ring.

The coefficient description of A_0 shows that it is exactly the unit ball for the restricted Gauss norm on A . Since this norm is multiplicative,

$$\|a^n\| = \|a\|^n.$$

It follows that a is power-bounded if and only if $\|a\| \leq 1$. Thus $A^\circ = A_0$, and the unit ball is bounded, so A is uniform.

To see that A is not noetherian, let $J = \ker(\iota_B) = Q^2C \subset A$. If J were generated by $a_1, \dots, a_r$, write $b_i \in L$ for the coefficient of Q^2 in a_i . Now, for every $\ell \in L$, we can write

$$Q^2\ell = \sum_i x_i a_i, \quad x_i \in A,$$

and write c_i for the constant coefficient of x_i as a series in Q . By (3), we have $c_i \in k\langle W \rangle$. Since $a_i \in Q^2C$, its coefficients in degrees 0 and 1 vanish. The coefficient of Q^2 in $x_i a_i$ is therefore exactly $c_i b_i$. Comparing the coefficients of Q^2 on both sides gives

$$\ell = \sum_i c_i b_i, \quad c_i \in k\langle W \rangle.$$

Thus $b_1, \dots, b_r$ would generate L as a $k\langle W \rangle$ -module. This is impossible. Indeed, it would make W^{-1} integral over $k\langle W \rangle$. Multiplying a monic relation for W^{-1} of degree n by W^n would put 1 in the proper ideal $(W) \subset k\langle W \rangle$. Thus J is not finitely generated, and A is not noetherian. $\square$

4. A RATIONAL LOCALISATION WHICH IS NOT UNIFORM

We next compute the rational chart which witnesses the failure of stable uniformity.

Consider the datum

$$\alpha = (W; \varpi).$$

The ideal (ϖ, W) is open, since ϖ is a unit in A . Let

$$A_\alpha = A \left\langle \frac{W}{\varpi} \right\rangle$$

be the corresponding rational localisation on which $|W| \leq |\varpi|$, and write $X = W/\varpi$. Since ϖ is already invertible in A , the algebraic localisation does not change the ring. More explicitly, A_α is obtained by taking the ϖ -adic completion of $A_0[X] \subset A$ and then inverting ϖ .

Proposition 4.1. *There is a canonical isomorphism of complete topological rings*

$$A_\alpha \cong k\langle X, Q \rangle / (Q^2), \quad W \mapsto \varpi X, \quad Q \mapsto Q.$$

Proof. Put

$$T_0 = k^\circ\langle X, Q \rangle / (Q^2), \quad T = T_0[1/\varpi].$$

Let $\rho : A \rightarrow A_\alpha$ be the canonical map. We first show that

$$Q^2C \subseteq \ker(\rho). \tag{5}$$

Take $y \in Q^2C$, and choose $r \geq 0$ such that $\varpi^r y \in Q^2C_0$. For every $n \geq 0$, the coefficient description of A_0 gives

$$W^{-n}\varpi^r y \in A_0.$$

Multiplication by W^{-n} only introduces negative powers of W in terms of Q -degree at least two. In $A_0[X]$, where $W = \varpi X$, we therefore have

$$\varpi^r y = W^n(W^{-n}\varpi^r y) = \varpi^n X^n(W^{-n}\varpi^r y) \in \varpi^n A_0[X]$$

for every n . Hence $\varpi^r y$ maps to zero in $\widehat{A_0[X]}$. Since ϖ is invertible in A_α , this proves (5).

We now construct the isomorphism. Recall that the map $\iota_B : A \rightarrow B$ from the defining pullback is given by

$$f_0(W) + Qf_1(W) + Q^2h \mapsto f_0(W) + Qf_1(W);$$

it simply discards the terms divisible by Q^2 . The substitution $W \mapsto \varpi X$, $Q \mapsto Q$ defines a continuous homomorphism $B \rightarrow T$. Their composite is

$$\theta : A \rightarrow T, \quad f_0(W) + Qf_1(W) + Q^2h \mapsto f_0(\varpi X) + Qf_1(\varpi X).$$

It sends A_0 into T_0 and sends W/ϖ to X . It therefore sends $A_0[X]$ into T_0 and extends, first to the completions and then after inverting ϖ , to a continuous homomorphism

$$\Phi : A_\alpha \rightarrow T.$$

Set

$$\overline{X} = \rho(W)/\varpi, \quad \overline{Q} = \rho(Q).$$

Both elements are power-bounded: they belong to the completed ring of definition coming from $A_0[X]$. Moreover, $\overline{Q}^2 = 0$ by (5). Evaluation at $\overline{X}$ therefore gives a continuous homomorphism

$$\Psi : T \rightarrow A_\alpha, \quad f + Qg \mapsto f(\overline{X}) + g(\overline{X})\overline{Q}.$$

It remains to check the two composites. Write $a = f_0(W) + Qf_1(W) + y \in A$, with $y \in Q^2C$. Then

$$(\Psi \circ \Phi)(\rho(a)) = f_0(\varpi \overline{X}) + \overline{Q}f_1(\varpi \overline{X}) = \rho(f_0(W) + Qf_1(W)) = \rho(a),$$

where the last equality uses (5). The image of A is dense in A_α , so continuity gives $\Psi \circ \Phi = 1$. Conversely, $\Phi \circ \Psi$ fixes k , X and Q , and hence is the identity on the dense subring $k[X, Q]/(Q^2)$ of T . It follows by continuity that $\Phi \circ \Psi = 1$. Thus Φ and Ψ are inverse continuous homomorphisms. $\square$

Proposition 4.2. *The ring A is not stably uniform.*

Proof. In $T = k\langle X, Q \rangle / (Q^2)$, every element of the line kQ is nilpotent and hence power-bounded. This line is not bounded with respect to the ring of definition T_0 : for every $r \geq 0$,

$$\varpi^r(\varpi^{-(r+1)}Q) = \varpi^{-1}Q \notin T_0.$$

Thus T is not uniform. By Proposition 4.1, A_α is not uniform. $\square$

5. LOCALISATION OF THE MILNOR SQUARE

In this section we prove that rational localisation preserves the defining pullback. We first record the strict Koszul estimate needed for the proof.

5.1. Koszul complexes for rational localisations. Let E be one of B, C, D , equipped with its coefficientwise Gauss norm, and let E_0 denote its unit ball. Let $f_1, \dots, f_m, g \in E_0$ generate the unit ideal in E . Put

$$T_E = E\langle T_1, \dots, T_m \rangle, \quad T_{E,0} = E_0\langle T_1, \dots, T_m \rangle,$$

and set $r_i = gT_i - f_i$.

We give T_E the topology for which $\{\varpi^n T_{E,0}\}_{n \geq 0}$ is a basis of neighbourhoods of zero. In the standard exterior basis, $\bigwedge^q T_E^m$ has the finite product topology, and each image has the subspace topology. Thus a differential is strict precisely when it is open onto its image.

Applied to $E \in \{B, C, D\}$, the strict Koszul regularity theorem of Bambozzi–Kremnizer [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13] gives the exactness, strictness, and closed-image assertions below. We include the direct proof followed in the formalisation because the two bounds (6) and (7) are not stated there and are needed below.

Lemma 5.1. *The Koszul complex*

$$0 \longrightarrow \bigwedge^m T_E^m \longrightarrow \cdots \longrightarrow \bigwedge^2 T_E^m \xrightarrow{d_{2,E}} T_E^m \xrightarrow{d_{1,E}} T_E$$

is exact in positive degrees. Every differential is continuous and strict onto its image, and every image is closed. In particular, if

$$I_E = (r_1, \dots, r_m) = \text{im}(d_{1,E}) \subset T_E, \quad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i r_i,$$

then I_E is closed. Moreover, there exists $h_E \in \mathbb{Z}_{\geq 0}$ such that

$$\varpi^{h_E}(I_E \cap T_{E,0}) \subset d_{1,E}(T_{E,0}^m). \quad (6)$$

There also exists $z_E \in \mathbb{Z}_{\geq 0}$ such that

$$\varpi^{z_E}(\ker(d_{1,E}) \cap T_{E,0}^m) \subset d_{2,E}(\bigwedge^2 T_{E,0}^m). \quad (7)$$

Proof. Over the polynomial ring $E[T_1, \dots, T_m]$, the sequence $gT_i - f_i$ has Koszul complex exact in positive degrees. Indeed, this may be checked after localising at a prime. If one of the $gT_i - f_i$ is not in the prime, it is a unit after localisation and the Koszul complex is contractible. If all of them lie in the prime, then g does not: otherwise all the f_i would also lie in the prime, contrary to $(g, f_1, \dots, f_m) = E$. Thus g is invertible there and

$$gT_i - f_i = g(T_i - f_i/g).$$

The elements $T_i - f_i/g$ form a regular sequence, since they are monic linear polynomials in distinct variables. This proves the assertion over the polynomial ring.

The map

$$E[T_1, \dots, T_m] \longrightarrow T_E$$

is flat. In fact, E_0 is noetherian for $E = B, C, D$, so $E_0[T_1, \dots, T_m]$ is noetherian and its ϖ -adic completion is flat over it [20, Tag 00MB]. Inverting ϖ gives the displayed flat map. Tensoring the polynomial Koszul complex with T_E , and using the standard base-change identification of its finite free terms, proves positive-degree exactness. Notice that this does not assert exactness in degree zero.

Each differential is continuous because its coordinates are finite sums of multiplication maps. The image of $d_{1,E}$ is closed by the noetherian closed-ideal theorem. In higher degree, exactness identifies an image with the kernel of the next continuous differential, and hence that image is closed. The nonarchimedean open mapping theorem therefore shows that, for each differential $d_{q,E}$, there is a constant $C_{q,E} \geq 1$ such that every $y \in \text{im}(d_{q,E})$ has a preimage x satisfying

$$d_{q,E}(x) = y, \quad \|x\| \leq C_{q,E}\|y\|.$$

This also proves that each differential is strict onto its image.

Choose $h_E \geq 0$ such that $|\varpi|^{h_E} C_{1,E} \leq 1$. If $x \in I_E \cap T_{E,0}$, choose $u \in T_E^m$ with

$$d_{1,E}(u) = x, \quad \|u\| \leq C_{1,E}\|x\| \leq C_{1,E}.$$

Then $\varpi^{h_E} u \in T_{E,0}^m$ and $d_{1,E}(\varpi^{h_E} u) = \varpi^{h_E} x$, which proves (6). Similarly, choose $z_E \geq 0$ such that $|\varpi|^{z_E} C_{2,E} \leq 1$. If $u \in \ker(d_{1,E}) \cap T_{E,0}^m$, choose v with $d_{2,E}(v) = u$ and $\|v\| \leq C_{2,E}\|u\|$. Then $\varpi^{z_E} v \in \bigwedge^2 T_{E,0}^m$, proving (7). $\square$

5.2. The defining ideals under pullback. Fix a rational datum

$$\alpha = (f_1, \dots, f_m; g)$$

in A . Multiplying all entries by a common power of ϖ , we assume $f_1, \dots, f_m, g \in A_0$. Put

$$r_i = gT_i - f_i \in T_A,$$

and use the same notation for its image in T_B, T_C , and T_D . For $E \in \{A, B, C, D\}$ put

$$I_E = (r_1, \dots, r_m) = \text{im}(d_{1,E}) \subset T_E.$$

The exact integral row (2) and the coefficientwise section of $C_0 \rightarrow D_0$ give an exact sequence

$$0 \longrightarrow T_{A,0} \longrightarrow T_{B,0} \oplus T_{C,0} \longrightarrow T_{D,0} \longrightarrow 0. \quad (8)$$

The map $T_{C,0} \rightarrow T_{D,0}$ is surjective, as are the induced maps on finite free modules and their exterior powers. Exactness is coefficientwise: a compatible restricted coefficient pair lifts to A_0 , and the lifted coefficients still tend to zero ϖ -adically.

Lemma 5.2. *The ideal I_A is closed in T_A , and the canonical map*

$$I_A \longrightarrow I_B \times_{I_D} I_C$$

is an isomorphism. Give each I_E the subspace topology inherited from T_E , and give $I_B \oplus I_C$ the max norm. Then the sequence of topological k -vector spaces

$$0 \longrightarrow I_A \longrightarrow I_B \oplus I_C \longrightarrow I_D \longrightarrow 0 \quad (9)$$

is strict exact. Here the maps are

$$x \longmapsto (x_B, x_C), \quad (x_B, x_C) \longmapsto (x_B)_D - (x_C)_D.$$

Proof. For $E = B, C, D$, let $C_{1,E}$ be as in the proof of Lemma 5.1, and let $C_{2,D}$ be the corresponding constant for $d_{2,D}$. Take $(x_B, x_C) \in I_B \times_{I_D} I_C$, and choose $u_B \in T_B^m$ and $u_C \in T_C^m$ such that

$$\begin{aligned} d_{1,B}(u_B) &= x_B, & \|u_B\| &\leq C_{1,B}\|x_B\|, \\ d_{1,C}(u_C) &= x_C, & \|u_C\| &\leq C_{1,C}\|x_C\|. \end{aligned}$$

In T_D^m the difference

$$w = (u_B)_D - (u_C)_D$$

belongs to $\ker(d_{1,D})$. Choose $v_D \in \bigwedge^2 T_D^m$ such that

$$d_{2,D}(v_D) = w, \quad \|v_D\| \leq C_{2,D}\|w\|.$$

The norm-preserving coefficientwise section $T_D \rightarrow T_C$ induces a norm-preserving section on second exterior powers; use it to obtain $v_C \in \bigwedge^2 T_C^m$ mapping to v_D . Replacing u_C by

$$u'_C = u_C + d_{2,C}(v_C)$$

does not change $d_{1,C}(u_C)$, and the images of u_B and u'_C in T_D^m now agree. Exactness of (8), applied coefficientwise, gives $u_A \in T_A^m$ mapping to this pair. Then $d_{1,A}(u_A)$ maps to (x_B, x_C) .

These bounds, continuity of $d_{2,C}$, and the norm-preserving coefficientwise section give a constant C , independent of (x_B, x_C) , for which $x_A = d_{1,A}(u_A)$ satisfies

$$\|x_A\| \leq C \max\{\|x_B\|, \|x_C\|\}.$$

The map $T_A \rightarrow T_B \oplus T_C$ is injective, so this proves both the algebraic pullback assertion and the boundedness of its inverse.

It remains to see that I_A is closed. The ideals I_B and I_C are closed by Lemma 5.1. If x lies in the closure of I_A , its images lie in I_B and I_C . The pullback isomorphism just proved gives $x_A \in I_A$ with the same two images as x . Injectivity of $T_A \rightarrow T_B \oplus T_C$ gives $x = x_A$, as required.

Finally, for $y \in I_D$ choose $u_D \in T_D^m$ such that

$$d_{1,D}(u_D) = y, \quad \|u_D\| \leq C_{1,D}\|y\|.$$

Apply the norm-preserving section $T_D^m \rightarrow T_C^m$ and then $d_{1,C}$. This gives $x_C \in I_C$ mapping to y , with $\|x_C\|$ bounded by a fixed multiple of $\|y\|$. Thus $I_B \oplus I_C \rightarrow I_D$ is a strict surjection. Its kernel is I_A by the pullback assertion, and the preceding bound for the inverse $I_B \times_{I_D} I_C \rightarrow I_A$ makes the inclusion on the left strict. This proves (9). $\square$

Proposition 5.3. *For every rational datum α in A , the canonical sequence*

$$0 \longrightarrow A_\alpha \longrightarrow B_\alpha \oplus C_\alpha \longrightarrow D_\alpha \longrightarrow 0$$

is strict exact, and $C_\alpha \rightarrow D_\alpha$ is a strict surjection. Equivalently,

$$A_\alpha \cong B_\alpha \times_{D_\alpha} C_\alpha.$$

The coefficient maps used in these identifications commute with rational restriction.

Proof. The ideals I_B, I_C, I_D are closed by Lemma 5.1, and I_A is closed by Lemma 5.2. Hence (1) gives

$$E_\alpha = T_E/I_E \quad (E = A, B, C, D).$$

We first prove algebraic exactness. If a representative $p \in T_A$ maps into both I_B and I_C , the pullback assertion in Lemma 5.2 gives an element of I_A with the same two images. Injectivity of $T_A \rightarrow T_B \oplus T_C$ then shows that $p \in I_A$. This gives injectivity on the left. Conversely, take compatible classes in T_B/I_B and T_C/I_C and choose representatives b and c . Their defect in T_D belongs to I_D . Lift this defect to an element of I_C using the strict surjection in (9), and add it to c . The corrected representatives agree in T_D , so (8) lifts them to T_A . This proves algebraic exactness.

For the topology on the left, let $x \in A_\alpha$ and put

$$M = \max\{\|x_B\|, \|x_C\|\}.$$

Choose representatives $b \in T_B$ and $c \in T_C$ whose norms are arbitrarily close to the quotient norms of x_B and x_C . Their defect lies in I_D . The norm estimate for $I_C \rightarrow I_D$ proved in Lemma 5.2 gives a constant C_0 and a correction $y \in I_C$ mapping to this defect such that

$$\|y\| \leq C_0 \|b_D - c_D\|.$$

Thus b and $c + y$ agree in T_D and lift to $p \in T_A$. The max-norm identity for the ambient pullback gives, after letting the error in the representatives tend to zero,

$$\|x\| \leq CM$$

for a constant C depending only on the datum. The two quotient maps are norm-nonincreasing, so this estimate says precisely that $A_\alpha \rightarrow B_\alpha \oplus C_\alpha$ is a topological embedding.

The coefficientwise section $T_D \rightarrow T_C$ shows directly that $C_\alpha \rightarrow D_\alpha$ is surjective. More precisely, choose a representative in T_D whose norm is arbitrarily close to the quotient norm of a given class in D_α , and apply the norm-preserving section. Its class in C_α has no larger norm. Thus the surjection is open; together with continuity, this says that it is strict.

All maps used above come from the coefficient maps $A \rightarrow B, C$ and $B, C \rightarrow D$. More generally, a homomorphism $E \rightarrow E'$ carrying a rational datum α to α' induces a coefficientwise map $T_E \rightarrow T_{E'}$ which sends the relations for α to those for α' . The pullback identifications therefore commute with these maps. Applying this to rational restriction, and then using presentation independence and transitivity, proves the asserted compatibility under refinement. $\square$

6. STRONG SHEAFINESS

Theorem 6.1. *The ring A is strongly sheafy. In particular, the Huber pair (A, A°) is sheafy.*

Proof. Give B, C, D their maximal rings of integral elements. The max/Gauss estimates (4) show that every structural map is norm-nonincreasing. Consequently it sends bounded subsets to bounded subsets; in particular, if x is power-bounded, then the set of powers of its image is the image of the bounded set $\{x^n : n \geq 0\}$. The structural maps therefore send power-bounded elements to power-bounded elements and define morphisms of Huber pairs.

Each of B, C, D is a classical strongly noetherian affinoid k -algebra and is therefore sheafy by Huber's theorem [11, Theorem 2.2]. Let $U = \bigcup_i U_i$ in $\mathrm{Spa}(A, A^\circ)$. Push the datum for U and the data for the U_i to B, C, D . If a section over U restricts to zero on every U_i , its images over B and C vanish by separatedness there. Injectivity of the localised Milnor row from Proposition 5.3 then shows that the original section is zero.

Now take a matching family over the U_i . Its images glue over B and C . The two glued sections have the same image over D , since their restrictions agree on the pushed cover and the presheaf for D is separated. Exactness of the localised Milnor row over U gives a section over A , and injectivity of the corresponding rows over the U_i shows that it restricts to the given family.

We must also check the topological part of the sheaf condition. The image of

$$\mathcal{O}(U) \longrightarrow \prod_i \mathcal{O}(U_i)$$

is exactly the subspace of tuples whose two restrictions to every $U_i \cap U_j$ agree. This subspace is closed: it is the common zero locus of the continuous difference maps to $\mathcal{O}(U_i \cap U_j)$, and hence is a Banach k -space. The restriction map is a continuous bijection from the Banach k -space $\mathcal{O}(U)$ to this closed subspace. The nonarchimedean open mapping theorem therefore makes it a homeomorphism. This proves the sheaf condition on finite rational covers. The sheaf-on-a-rational-basis theorem [20, Tag 009O] then gives the all-open topological sheaf condition for (A, A°) .

The same proof applies after adjoining $V_1, \dots, V_s$. The split Milnor row extends coefficientwise, the three other vertices remain strongly noetherian affinoid k -algebras, and the Koszul argument above is unchanged. Thus $A\langle V_1, \dots, V_s \rangle$ is sheafy for every $s \geq 0$. $\square$

Proof of Theorem 1.1. Proposition 3.1 proves that A is complete, uniform, non-noetherian, and a domain. Strong sheafiness is Theorem 6.1, and failure of stable uniformity is Proposition 4.2. $\square$

7. PROBLEMS 24 AND 28

Problem 24 of the *Nonarchimedean Scottish Book* asks whether the map of rings underlying a rational localisation of Tate Huber pairs must be flat. Problem 28 asks whether a non-zero-divisor which generates a closed ideal can restrict to zero on a nonempty rational subspace [19, Problems 24 and 28]. The same rational datum, together with multiplicativity of the Gauss norm, answers both questions. We keep the notation of the introduction. In fact, the argument only uses that k is complete and ultrametric and that $0 < |\varpi| < 1$; neither conclusion uses sheafiness.

Proposition 7.1. *Let*

$$U = \{x \in \mathrm{Spa}(A, A^\circ) : |W(x)| \leq |\varpi(x)|\}.$$

Then U is nonempty, the rational localisation

$$A \longrightarrow \mathcal{O}(U) = A \left\langle \frac{W}{\varpi} \right\rangle$$

is not flat, and multiplication by Q^2 is a strict inclusion on A even though Q^2 maps to zero in $\mathcal{O}(U)$.

Proof. The composite

$$A \longrightarrow B = k\langle W, Q \rangle / (Q^2) \longrightarrow k, \quad f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(0),$$

is norm-nonincreasing. It therefore sends A° into k° , and pulling back the norm valuation on k gives a continuous valuation belonging to $\mathrm{Spa}(A, A^\circ)$. At this point W has value zero and ϖ has nonzero value, so the point lies in U .

Since the restricted Gauss norm on A is multiplicative and $\|Q\| = 1$, for every $a \in A$ we have

$$\|Q^2a\| = \|a\|.$$

Thus multiplication by Q^2 is an isometry. Its map onto its image, with the subspace topology, is consequently a homeomorphism, so the inclusion is strict. Moreover, an isometric image of the complete space A is closed in A . Since $\|Q^2\| = 1$, the element Q^2 is nonzero; multiplication by it is therefore injective because A is a domain.

On the other hand, the calculation (5) says directly that the canonical map to the completed rational localisation kills Q^2 . The isomorphism of Proposition 4.1 identifies this localisation with the nonzero ring

$$A \longrightarrow k\langle X, Q \rangle / (Q^2), \quad W \longmapsto \varpi X, \quad Q \longmapsto Q.$$

If the canonical map were flat, the non-zero-divisor Q^2 would still act injectively on $\mathcal{O}(U)$; equivalently, tensoring the injection

$$0 \longrightarrow A \xrightarrow{\cdot Q^2} A$$

with $\mathcal{O}(U)$ would make multiplication by the image of Q^2 injective. Its image is zero, however, so this multiplication map is zero on the nonzero ring $\mathcal{O}(U)$, a contradiction. $\square$

8. A SECOND EXAMPLE

In the finite-jet construction, the bad rational localisation is non-reduced. In our second example it is an integral domain; the loss of uniformity is instead caused by an unbounded family of non-nilpotent power-bounded elements.

Theorem 8.1. *There is a complete Tate k -algebra $\mathcal{A}$ with the following properties.*

- (1) $\mathcal{A}$ is a uniform, non-noetherian integral domain and $\mathcal{A}^\circ = \mathcal{A}_0$, where $\mathcal{A}_0$ is defined below.
- (2) The ring $\mathcal{A}$ is strongly sheafy; in particular, the Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.
- (3) The rational localisation

$$\mathcal{B} = \mathcal{A} \left\langle \frac{W}{\varpi} \right\rangle$$

is an integral domain but is not uniform.

Consequently $\mathcal{A}$ is not stably uniform. In particular, failure of stable uniformity need not be caused by a nilpotent in the bad rational localisation.

8.1. The weighted-parity algebra. We write $\mathbb{N} = \{0, 1, 2, \dots\}$ and put $I = \mathbb{N}_{>0}$. For a finite-support multi-index $\nu = (\nu_n)_{n \in I} \in \mathbb{N}^{(I)}$, put

$$\omega(\nu) = \sum_{\substack{n \geq 1 \\ \nu_n \text{ odd}}} n. \quad (10)$$

Consider the submonoid

$$S = \left\{ (a, \nu) \in \mathbb{N} \times \mathbb{N}^{(I)} : a \geq \omega(\nu) \right\}. \quad (11)$$

The inequality

$$\omega(\nu + \mu) \leq \omega(\nu) + \omega(\mu) \quad (12)$$

shows that S is closed under addition. It is locally finite: a fixed element of S has only finitely many decompositions as a sum of two elements of S .

Let $k[S]$ be the monoid algebra with basis $W^a U^\nu$, where $U^\nu = \prod_n U_n^{\nu_n}$. Give it the Gauss norm

$$\left\| \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu \right\|_G = \sup_{a, \nu} |c_{a, \nu}|. \quad (13)$$

Its completion is

$$\mathcal{A} = \left\{ \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu : c_{a, \nu} \in k, \quad c_{a, \nu} \rightarrow 0 \right\}, \quad (14)$$

where convergence to zero is over the discrete index set S . Local finiteness makes the coefficientwise convolution finite, and the submultiplicativity of the Gauss norm on $k[S]$ extends multiplication continuously to $\mathcal{A}$. Put

$$\mathcal{A}_0 = \left\{ \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu \in \mathcal{A} : c_{a, \nu} \in k^\circ \right\}.$$

Then $\mathcal{A}_0$ is the closed unit ball for (13), it is ϖ -adically complete, and $\mathcal{A} = \mathcal{A}_0[1/\varpi]$.

Put

$$Y_n = W^n U_n, \quad Z_n = U_n^2,$$

and, for $N \in \mathbb{N}$, let $\mathcal{A}^{(N)} \subset \mathcal{A}$ be the closed subring supported on monomials involving only $W, U_1, \dots, U_N$. Thus $\mathcal{A}^{(0)} = k\langle W \rangle$. Let $\mathcal{T}^{(N)} \subset \mathcal{A}^{(N)}$ be the subring in which every U_n has even exponent.

Lemma 8.2 (Finite-variable subrings). *For every $N \in \mathbb{N}$, substitution $Z_n \mapsto U_n^2$ gives a norm-preserving ring isomorphism*

$$k\langle W, Z_1, \dots, Z_N \rangle \xrightarrow{\sim} \mathcal{T}^{(N)}.$$

For $N = 0$ this is the evident identification of both sides with $k\langle W \rangle$. Every $f \in \mathcal{A}^{(N)}$ can be written

$$f = \sum_{\epsilon \in \{0,1\}^N} f_\epsilon Y^\epsilon, \quad f_\epsilon \in \mathcal{T}^{(N)}, \quad Y^\epsilon = \prod_{n=1}^N Y_n^{\epsilon_n}. \quad (15)$$

Consequently $\mathcal{A}^{(N)}$ is finite over $\mathcal{T}^{(N)}$ and is a strongly noetherian Tate ring. The inclusions $\mathcal{A}^{(N)} \rightarrow \mathcal{A}^{(M)} \rightarrow \mathcal{A}$ for $N \leq M$ preserve the norm. Moreover, for every $f \in \mathcal{A}$ and $\ell \geq 0$ there are N and $f_N \in \mathcal{A}^{(N)}$ such that

$$\|f - f_N\|_G \leq |\varpi|^\ell \|f\|_G.$$

In particular,

$$\mathcal{A} = \bigcup_{N \geq 0} \widehat{\mathcal{A}^{(N)}}. \quad (16)$$

Proof. On even exponents, halving the U_n -coordinates identifies the support monoid with that of the Tate algebra $k\langle W, Z_1, \dots, Z_N \rangle$; the coefficient map and its inverse preserve the Gauss norm. Write each exponent uniquely as $\nu_n = 2q_n + \epsilon_n$, with $\epsilon_n \in \{0, 1\}$. Since $\omega(\nu) = \sum_{\epsilon_n=1} n$, every allowed monomial factors as

$$W^a U^\nu = W^{a-\omega(\nu)} \prod_{n=1}^N Z_n^{q_n} \prod_{n=1}^N Y_n^{\epsilon_n}.$$

Grouping the monomials with the same tuple $(\epsilon_1, \dots, \epsilon_N)$ gives (15). The finitely many Y^ϵ therefore generate $\mathcal{A}^{(N)}$ over $\mathcal{T}^{(N)}$. The same argument after adjoining any finite number of Tate variables gives strong noetherianity. The inclusions are coefficientwise and hence isometric. Finally, approximate f to the required precision by a finite sum of monomials and choose N so that only $U_1, \dots, U_N$ occur in that sum. This gives the stated estimate and (16). $\square$

Proposition 8.3. *The ring $\mathcal{A}$ is a complete uniform Tate k -algebra and an integral domain, with*

$$\mathcal{A}^\circ = \mathcal{A}_0.$$

Proof. The Gauss norm on the countable restricted Tate algebra $k\langle W, U_1, U_2, \dots \rangle$ is multiplicative. To see this, note first that only finitely many coefficients attain the nonzero supremum, since the coefficients tend to zero. Order the finite-support exponent vectors by comparing them at the largest index where they differ. This is an additive total order. For each of two nonzero series choose the largest norm-attaining exponent. In the coefficient of the sum of these two exponents, the product of the chosen coefficients strictly dominates every other convolution term. The ultrametric inequality therefore gives

$$\|fg\|_G = \|f\|_G \|g\|_G.$$

This is the countable restricted-series argument used in the formalisation, whose Lean implementation uses William Coram's restricted-power-series code [6]. The support algebra $\mathcal{A}$ is a

closed isometric subring of this Tate algebra, so its Gauss norm is multiplicative and it is a domain.

If $x \in \mathcal{A}_0$, all powers of x remain in $\mathcal{A}_0$. If $x \notin \mathcal{A}_0$, multiplicativity gives $\|x^m\|_G = \|x\|_G^m$, which is unbounded. Hence $\mathcal{A}^\circ = \mathcal{A}_0$. Completeness and the Tate property follow from (14) and the topologically nilpotent unit ϖ . $\square$

Proposition 8.4. *The ring $\mathcal{A}$ is not noetherian.*

Proof. For $m \geq 1$, let

$$I_m = (Z_1, \dots, Z_m) \subset \mathcal{A}.$$

Define a norm-nonincreasing homomorphism

$$\psi_m : \mathcal{A} \longrightarrow k\langle T \rangle,$$

by retaining only the coefficients of the monomials U_{m+1}^{2q} and sending that monomial to T^q . The support condition shows directly that this is multiplicative: a product can contribute such a monomial only when both factors contribute monomials of the same form. Thus ψ_m sends W and every Y_j to zero, sends Z_{m+1} to T , and sends every other Z_j to zero. It kills I_m but not Z_{m+1} . Therefore

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \dots,$$

so $\mathcal{A}$ is not noetherian. $\square$

8.2. Small perturbations of rational data.

Lemma 8.5 (Small perturbation lemma). *Let (E, E^+) be a complete Tate Huber pair and let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum in E . There is a neighbourhood V of zero in E such that, whenever*

$$g' - g, f'_1 - f_1, \dots, f'_m - f_m \in V,$$

the tuple $\alpha' = (f'_1, \dots, f'_m; g')$ is a rational datum and α and α' define the same rational subset of $\mathrm{Spa}(E, E^+)$. Their completed rational localisations are canonically bicontinuously isomorphic over E .

Proof. Apply [10, Lemma 3.10] after including g among the numerators; see also [13, Remark 2.4.7]. The assertion about completed rational localisations follows from their universal property [11, Proposition 1.3]. $\square$

8.3. Rational localisations defined over finitely many variables. Fix $N \in \mathbb{N}$. Let $\mu = (\mu_n)_{n > N}$ be a finitely supported family of nonnegative integers and put

$$e_\mu = W^{\omega(\mu)} U^\mu,$$

where μ is extended by zero for $n \leq N$. Every $f \in \mathcal{A}$ has a unique convergent expression

$$f = \sum_{\mu} f_{\mu} e_{\mu}, \quad f_{\mu} \in \mathcal{A}^{(N)}, \quad \|f_{\mu}\|_G \longrightarrow 0, \quad (17)$$

and

$$\|f\|_G = \sup_{\mu} \|f_{\mu}\|_G.$$

Multiplication is determined by

$$e_{\mu} e_{\lambda} = W^{\omega(\mu) + \omega(\lambda) - \omega(\mu + \lambda)} e_{\mu + \lambda}. \quad (18)$$

The map

$$\rho_N : \mathcal{A} \longrightarrow \mathcal{A}^{(N)}, \quad \sum_{\mu} f_{\mu} e_{\mu} \longmapsto f_0,$$

is a norm-nonincreasing ring retraction.

Proposition 8.6 (Coefficientwise localisation). *Let α be a rational datum whose entries lie in $\mathcal{A}^{(N)}$, and put $P = (\mathcal{A}^{(N)})_\alpha$. There is a bicontinuous ring isomorphism*

$$\mathcal{A}_\alpha \xrightarrow{\sim} \left\{ \sum_{\mu} p_{\mu} e_{\mu} : p_{\mu} \in P, \quad \|p_{\mu}\| \rightarrow 0 \right\}.$$

The ring on the right has the supremum norm, and its multiplication is given by (18), with W replaced by its image in P . If a rational refinement is also represented by data in $\mathcal{A}^{(N)}$ and $P \rightarrow P'$ is the corresponding restriction map, then the induced restriction sends

$$\sum_{\mu} p_{\mu} e_{\mu} \mapsto \sum_{\mu} (p_{\mu}|_{P'}) e_{\mu}.$$

Proof. Apply the canonical map $\mathcal{A}^{(N)} \rightarrow P$ to every coefficient in (17). This gives a continuous homomorphism

$$\Phi_0 : \mathcal{A} \rightarrow \left\{ \sum_{\mu} p_{\mu} e_{\mu} : p_{\mu} \in P, \quad \|p_{\mu}\| \rightarrow 0 \right\}.$$

In the target, the image of g is a unit and each f_i/g is of norm at most 1, by the defining property of P . Thus Φ_0 extends continuously to $\mathcal{A}[1/g]$ for the rational-localisation topology, and then to the completion:

$$\Phi : \mathcal{A}_\alpha \rightarrow \left\{ \sum_{\mu} p_{\mu} e_{\mu} : p_{\mu} \in P, \quad \|p_{\mu}\| \rightarrow 0 \right\}.$$

In the other direction, the inclusion $\mathcal{A}^{(N)} \rightarrow \mathcal{A}$ gives a continuous homomorphism $P \rightarrow \mathcal{A}_\alpha$. Each e_{μ} lies in $\mathcal{A}_0$, and the canonical map to a rational localisation is bounded. Hence the family $(e_{\mu})_{\mu}$ is bounded in $\mathcal{A}_\alpha$. Apply the homomorphism to every coefficient. If $\|p_{\mu}\| \rightarrow 0$, the series

$$\sum_{\mu} p_{\mu} e_{\mu}$$

converges in $\mathcal{A}_\alpha$. The resulting map Ψ is continuous, and (18) shows that it is multiplicative.

The composite $\Phi\Psi$ fixes every term $p e_{\mu}$, and hence every finite sum of such terms. These finite sums are dense. The other composite fixes the dense image of $\mathcal{A}[1/g]$. Continuity therefore gives $\Phi\Psi = 1$ and $\Psi\Phi = 1$. This proves the claimed isomorphism.

If $P \rightarrow P'$ comes from a rational refinement, all the maps just used commute with this restriction map. On a finite sum, restriction simply applies $P \rightarrow P'$ to each coefficient; continuity gives the stated formula for every convergent sum. $\square$

Corollary 8.7 (Finite-variable presentation). *Every rational localisation of $\mathcal{A}$ is bicontinuously isomorphic to a ring of the form*

$$\left\{ \sum_{\mu} p_{\mu} e_{\mu} : p_{\mu} \in P, \quad \|p_{\mu}\| \rightarrow 0 \right\},$$

where P is a rational localisation of one of the strongly noetherian Tate rings $\mathcal{A}^{(N)}$.

Proof. Start with a rational datum $\alpha = (f_1, \dots, f_m; g)$ in $\mathcal{A}$. Choose a scalar $t \in k^\times$ with $0 < |t| < 1$ and small enough that $tg, tf_1, \dots, tf_m$ lie in $\mathcal{A}_0$. Multiplying the datum by t does not change the rational subset or its localisation, so we may assume that $g, f_1, \dots, f_m$

lie in $\mathcal{A}_0$. Choose a neighbourhood V as in Lemma 8.5. By (16), for some N there are $g', f'_1, \dots, f'_m \in \mathcal{A}^{(N)} \cap \mathcal{A}_0$ such that

$$g' - g, f'_1 - f_1, \dots, f'_m - f_m \in V.$$

By Lemma 8.5, the datum $\alpha' = (f'_1, \dots, f'_m; g')$ gives the same rational subset and the same completed localisation. It is rational in $\mathcal{A}$, so applying ρ_N to a Bezout identity for its entries shows that it is already a rational datum in $\mathcal{A}^{(N)}$. Now apply Proposition 8.6. $\square$

8.4. A reduced rational chart which is not uniform. The ideal (W, ϖ) is open because ϖ is a unit of $\mathcal{A}$. Define the complete k° -algebra

$$\mathcal{B}_0 = \left\{ \sum_{\nu \in \mathbb{N}^{(I)}, d \geq \omega(\nu)} b_{d,\nu} X^d U^\nu : b_{d,\nu} \in \varpi^{\omega(\nu)} k^\circ, \quad \varpi^{-\omega(\nu)} b_{d,\nu} \longrightarrow 0 \right\}, \quad (19)$$

with norm

$$\left\| \sum b_{d,\nu} X^d U^\nu \right\|_{\mathcal{B}} = \sup_{d,\nu} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|. \quad (20)$$

The subadditivity (12) shows that $\mathcal{B}_0$ is a ring. Put $\mathcal{B} = \mathcal{B}_0[1/\varpi]$.

Proposition 8.8. *There is a bicontinuous ring isomorphism*

$$\mathcal{A} \left\langle \frac{W}{\varpi} \right\rangle \xrightarrow{\sim} \mathcal{B}, \quad W \longmapsto \varpi X.$$

Proof. The datum $(W; \varpi)$ lies in $\mathcal{A}^{(0)} = k\langle W \rangle$. Put

$$P = \mathcal{A}^{(0)} \langle T \rangle / (\overline{\varpi T - W}).$$

Evaluation at $W = \varpi X$ and $T = X$ gives a continuous map $P \rightarrow k\langle X \rangle$. Conversely, evaluation of $k\langle X \rangle$ at the image of T gives a continuous map in the other direction. The two maps are inverse by polynomial density, and the norm estimates in the two universal properties show that the isomorphism is isometric.

Apply Proposition 8.6 with $N = 0$. It describes the rational chart as the ring of sums

$$\sum_{\nu} p_{\nu} e_{\nu}, \quad p_{\nu} \in P, \quad \|p_{\nu}\| \longrightarrow 0.$$

Under $P \cong k\langle X \rangle$, this is precisely the model (19). Its sup norm is (20), because the term indexed by ν is multiplied by $\varpi^{\omega(\nu)}$. The map from $\mathcal{A}^{(0)}$ sends W to ϖX . $\square$

Proposition 8.9. *The ring $\mathcal{B}$ is an integral domain but is not uniform.*

Proof. Put $R = k\langle X \rangle$. Here

$$R[[U_1, U_2, \dots]] = \left\{ \sum_{\nu \in \mathbb{N}^{(I)}} a_{\nu} U^{\nu} : a_{\nu} \in R \right\}$$

denotes the ring of all coefficient families indexed by finite-support multi-indices. Multiplication is convolution. For a fixed ν there are only finitely many decompositions $\nu = \lambda + \mu$, so every coefficient of a product is a finite sum. Order $\mathbb{N}^{(I)}$ by comparing two vectors at the largest index where they differ. This is an additive well-order. If two series are nonzero, the coefficient at the sum of their least nonzero exponents is the product of the corresponding coefficients and is nonzero. Thus $R[[U_1, U_2, \dots]]$ is a domain.

The coefficient description gives a homomorphism

$$\mathcal{B} \longrightarrow R[[U_1, U_2, \dots]], \quad \sum_{\nu} p_{\nu} e_{\nu} \longmapsto \sum_{\nu} (\varpi X)^{\omega(\nu)} p_{\nu}(X) U^{\nu}.$$

It is a homomorphism by (18). If its image is zero, then

$$(\varpi X)^{\omega(\nu)} p_{\nu}(X) = 0$$

for every ν . Since $k\langle X \rangle$ is a domain, every p_{ν} is zero. The map is therefore injective, and hence $\mathcal{B}$ is a domain.

Inside $\mathcal{A}$ write $Y_n = W^n U_n$ and put

$$T_n = \varpi^{-n} Y_n = X^n U_n \in \mathcal{B}.$$

Formula (20) gives

$$\|T_n\|_{\mathcal{B}} = |\varpi|^{-n}, \quad \|T_n^2\|_{\mathcal{B}} = 1.$$

The even powers of T_n have norm at most 1, and the odd powers have norm at most $\|T_n\|_{\mathcal{B}}$. Thus every T_n is power-bounded. The family $(T_n)_{n \geq 1}$ is not bounded, since its norms $|\varpi|^{-n}$ are unbounded. Hence $\mathcal{B}^{\circ}$ is unbounded and $\mathcal{B}$ is not uniform. $\square$

8.5. Strong sheafiness.

Theorem 8.10. *The ring $\mathcal{A}$ is strongly sheafy. In particular, the Huber pair $(\mathcal{A}, \mathcal{A}^{\circ})$ is sheafy.*

Proof. We first work with a finite rational cover $U = \bigcup_i U_i$. Choose one N large enough to approximate the datum for U and the data for all the U_i in $\mathcal{A}^{(N)}$. The standard combined data for every pairwise intersection $U_i \cap U_j$ then also lie in $\mathcal{A}^{(N)}$. The small perturbation lemma does not change the rational domains, and Proposition 8.6 gives compatible identifications

$$\begin{aligned} \mathcal{O}_X(U) &\cong \left\{ \sum_{\mu} p_{\mu} e_{\mu} : p_{\mu} \in P, \|p_{\mu}\| \rightarrow 0 \right\}, \\ \mathcal{O}_X(U_i) &\cong \left\{ \sum_{\mu} p_{i,\mu} e_{\mu} : p_{i,\mu} \in P_i, \|p_{i,\mu}\| \rightarrow 0 \right\}, \end{aligned}$$

with coefficientwise restriction maps. Here P is a rational localisation of the strongly noetherian Tate ring $\mathcal{A}^{(N)}$, and the P_i are rational localisations corresponding to the U_i . Let V and V_i be the rational subsets of

$$\mathrm{Spa}(\mathcal{A}^{(N)}, (\mathcal{A}^{(N)})^{\circ})$$

defined by the data giving P and P_i . Then $V = \bigcup_i V_i$. Indeed, if $v \in V$, the composite $v \circ \rho_N$ is a point of $\mathrm{Spa}(\mathcal{A}, \mathcal{A}^{\circ})$, since ρ_N sends power-bounded elements to power-bounded elements. For data in $\mathcal{A}^{(N)}$, membership in the corresponding rational subset is unchanged by this pullback because ρ_N is a retraction. Thus $v \circ \rho_N$ belongs to U , and hence to some U_i , which gives $v \in V_i$. We may therefore apply sheafiness of the strongly noetherian ring $\mathcal{A}^{(N)}$ to the cover $V = \bigcup_i V_i$.

Let $(x_i)_i$ be a matching family and write $x_i = \sum_{\mu} x_{i,\mu} e_{\mu}$. Since $\mathcal{A}^{(N)}$ is sheafy, the matching family $(x_{i,\mu})_i$ glues to a unique $q_{\mu} \in P$ for every μ . It remains to check that $\|q_{\mu}\| \rightarrow 0$. The restrictions of the q_{μ} to every P_i tend to zero because the coefficients of each x_i do. The product restriction map

$$P \longrightarrow \prod_i P_i$$

is a topological embedding, so it reflects convergence to zero. Hence $q_{\mu} \rightarrow 0$, and $\sum_{\mu} q_{\mu} e_{\mu}$ is the required section over U . The same coefficientwise argument proves separation.

The image of the product restriction map for $\mathcal{A}$ is therefore the subspace of tuples whose restrictions to every pairwise intersection agree. This subspace is closed: it is the common zero locus of the continuous difference maps, and hence is a Banach k -space. The restriction map is a continuous bijection from the Banach k -space $\mathcal{O}_X(U)$ to this closed subspace. The nonarchimedean open mapping theorem therefore makes it a homeomorphism. We have proved the topological sheaf condition on finite rational covers. The general rational-basis theorem upgrades this to the all-open structure presheaf, and its plus-ring independence gives the same conclusion for every ring of integral elements. In particular, $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.

For $s \geq 0$, the Tate extension $\mathcal{A}\langle V_1, \dots, V_s \rangle$ is bicontinuously isomorphic to the same weighted-parity construction with the first s weights equal to zero and the remaining weights shifted by s . The argument above applies to every weight function, so this Tate extension is sheafy. Hence $\mathcal{A}$ is strongly sheafy. $\square$

Proof of Theorem 8.1. The fact that $\mathcal{A}$ is a uniform domain is Proposition 8.3, and its failure to be noetherian is Proposition 8.4. Strong sheafiness is Theorem 8.10. Finally, Propositions 8.8 and 8.9 identify the rational chart $|W| \leq |\varpi|$ as an integral domain which is not uniform. $\square$

Remark 8.11 (Admissible blow-ups). There is a common formal interpretation of the two bad rational charts. Let R_0 denote A_0 in the finite-jet example and $\mathcal{A}_0$ in the weighted-parity example. The affine chart of the admissible blow-up of $\mathrm{Spf}(R_0)$ along (ϖ, W) on which the pullback of this ideal is generated by ϖ has ring

$$\widehat{R_0[W/\varpi]}.$$

Here $R_0[W/\varpi]$ means the subring of $R_0[1/\varpi]$ generated by R_0 and W/ϖ , and the hat denotes the ϖ -adic completion defined above. After inverting ϖ , its coordinate ring is the completed rational localisation $R_0[1/\varpi]\langle W/\varpi \rangle$, corresponding to $|W| \leq |\varpi|$. This description applies to our non-noetherian formal models because they are ϖ -adic and their ideal of definition (ϖ) is principal; see [7, Chapter 0, Section 7.1(c), and Chapter II, Section 1.1(b)].

For $R_0 = A_0$, the ring $A_0[W/\varpi]$ is a domain, but the calculation in the proof of Proposition 4.1 shows that

$$Q^2 \in \bigcap_{n \geq 1} \varpi^n A_0[W/\varpi].$$

This intersection is the kernel of the completion map. Thus Q^2 is killed, while Q survives, so this rational chart is non-reduced. The map from A to this completed rational localisation is also not flat by Proposition 7.1. For $R_0 = \mathcal{A}_0$, the completed chart remains a domain, but its generic fibre is not uniform because the power-bounded elements $T_n = X^n U_n$ form an unbounded family. Thus the two examples give different failures on the same kind of affine blow-up chart.

The proof of sheafiness for $\mathcal{A}$ uses the further fact that a rational datum can be approximated inside some strongly noetherian $\mathcal{A}^{(N)}$, after which localisation and restriction are computed coefficient by coefficient. The infinite monomial support and the unbounded power-bounded elements are in the tradition of the examples of Buzzard–Verberkmoes and Mihara; the approximation by the rings $\mathcal{A}^{(N)}$ is what makes sheafiness accessible here.

9. HOW THE EXAMPLES WERE FOUND

The authors had for some time been interested in finding a counterexample to Problem 7 of the *Nonarchimedean Scottish Book*. Our attempts had centred on the example of [4, Section 4.5, Proposition 17], which is uniform but not stably uniform. We believe that this example is in fact sheafy, but we have not found a proof.

In January 2026 we began to experiment with AI tools, initially ChatGPT 5.2 Pro. Buzzard–Verberkmoes reduce exactness for a two-member Laurent cover to a boundedness condition on the relevant rings of definition [4, Lemma 2]. For a uniform ring this follows from boundedness of the power-bounded elements [4, Lemma 3 and Corollary 4]. Their reduction from general rational covers to Laurent covers is given in [4, Lemma 8 and the proof of Theorem 7]. For sheafiness these checks must also be made after rational localisation. Their Theorem 7 cannot be applied directly to Proposition 17, since one of the rational localisations in that example is not uniform. Our plan was instead to ask ChatGPT to write down many explicit rational covers and check the required exactness directly for each one, in the hope that a pattern, and eventually a proof, would emerge.

After a couple of weeks ChatGPT 5.2 Pro claimed to have a complete proof. The authors then began the slow process of understanding this proof, only to discover, with the help of ChatGPT 5.3, that one of its arguments used the following assertion:

If $\phi : A \rightarrow B$ is a continuous homomorphism of Hausdorff topological abelian groups and A is complete, then $\phi(A)$ is closed in B .

This assertion is false, and its use completely broke the purported proof.

Motivated by this, we began in parallel to use Claude Code to formalise the parts of the theory of adic spaces needed here, including the theorem that strongly noetherian Tate rings are sheafy. The aim was a practical one: future AI-generated proofs could be formalised before the human authors had spent days understanding an argument containing a fatal error near its end.

We then returned to writing down covers and looking for a pattern. This continued through ChatGPT 5.3 and subsequent versions. When ChatGPT 5.6 Sol became available, we asked, almost in passing, whether it could prove the result we wanted or *find a new example*. It produced the examples studied here. By then the Lean development was far enough advanced that we could use Claude Code to formalise the arguments quickly in Lean and check that they were correct.

APPENDIX A. LEAN FORMALISATION AND VERIFICATION

This appendix records the Lean formalisation of the two examples. The finite-jet declarations are taken from AINTLIB commit `b007a4f3d`, and the weighted-parity declarations from commit `090a28921`. The comparison with Mathlib’s sheaf predicate is taken from commit `d92f96504`, and the results on Scottish Book Problems 24 and 28 from commit `01116aca6`. The strong-sheafiness statements are taken from commit `870d0eed2` [1]. For each example Lean proves the Tate structure and uniformity, proves that the ring is a domain but not noetherian, proves sheafiness and failure of stable uniformity, and verifies the rational chart calculation used in the proof.

All five pinned commits are public in AINTLIB. The interactive version of the paper includes each declaration used below, extracted from the appropriate commit, and its archival page links to the pinned source on GitHub. The verification report records the Lean and mathlib versions, the build, and the axiom checks.

The code excerpts below retain the mathematically meaningful bodies of definitions. Proof fields which only verify closure are omitted. For theorems we display only the statement and replace the proof by `...`; the links give the complete Lean declarations and their checked proofs.

A.1. The rational-cover predicate `IsSheafy`.

Definition A.1 (The rational-cover predicate). The proofs are carried out on finite rational covers, so the formalisation uses the class `IsSheafy`. We chose this formulation because

it is easier to work with for our purposes: it speaks directly about the completed rational localisations which occur in the proofs. The equivalences below show that this choice does not change the notion of sheafiness. The class is in the `ValuationSpectrum` namespace and is defined in `StructureSheaf.lean`. Here `RationalCoveringData A` consists of a rational datum for the base and finitely many data whose rational subsets cover it. The separate condition `C.IsRational` says that the numerator set in the base and in each member of the cover generates an open ideal. For a Tate ring this is equivalent to generating the unit ideal. Finally, `presheafValue D` is the completed rational localisation $A\langle T/s \rangle$ attached to D .

```
class IsSheafy (A : Type u) [CommRing A] [TopologicalSpace A]
  [IsTopologicalRing A] [inst1 : PlusSubring A]
  [inst2 : IsHuberRing A] [T2Space A] [NonarchimedeanRing A]
  [letI : UniformSpace A :=
    IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A]
  [IsRingOfIntegralElements (A+)] : Prop where
embedding : ∀ (C : RationalCoveringData A), C.IsRational →
  Topology.IsEmbedding (productRestrictionSub A C)
gluing : ∀ (C : RationalCoveringData A), C.IsRational →
  ∀ (f : ∀ (D : ↑C.covers), presheafValue D.1),
  (∀ (D1 D2 : ↑C.covers) (D3 : RationalLocData A)
    (h31 : rationalOpen D3.T D3.s ⊆
      rationalOpen D1.1.T D1.1.s)
    (h32 : rationalOpen D3.T D3.s ⊆
      rationalOpen D2.1.T D2.1.s),
    restrictionMap D1.1 D3 h31 (f D1) =
      restrictionMap D2.1 D3 h32 (f D2)) →
  ∃ x : presheafValue C.base, ∀ (D : ↑C.covers),
    restrictionMap C.base D.1 (C.hsubset D.1 D.2) x = f D
```

The two fields say that restriction to a finite rational cover is a topological embedding and that every matching family glues. Compatibility is written using common rational refinements; the exact-intersection theorem identifies this with the usual compatibility on pairwise intersections. Thus `IsSheafy A` is the topological sheaf condition on the rational basis.

A.2. Comparison with Mathlib's sheaf predicate.

Definition A.2 (Sections on arbitrary opens). Let $V \subseteq \text{Spa}(A, A^+)$ be open. The ring `limitSections V` consists of compatible sections over the rational subdomains contained in V , or equivalently their projective limit. If $W \subseteq V$, then `limitRestrict` is the map $\mathcal{O}_X(V) \rightarrow \mathcal{O}_X(W)$ obtained by reindexing this family. The corresponding all-open condition is `IsLimitSheaf`, from `SheafyPair.lean`.

The class `HasLocLiftPowerBounded` is used internally to construct the restriction maps between completed rational localisations. For an inclusion $R(D') \subseteq R(D)$, it says that the denominator of D becomes a unit on $R(D')$ and that the fractions t/s , for t in the numerator set of D , are power-bounded there. This class is supplied automatically for the complete Tate pairs considered here; it is not an additional hypothesis in our results.

```
structure IsLimitSheaf (A : Type u) [CommRing A] [TopologicalSpace A]
  [IsTopologicalRing A] [PlusSubring A] [IsHuberRing A]
  [HasLocLiftPowerBounded A] : Prop where
injective : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
  {U : ι → Opens ↑(Spa A A+)}
  (hle : ∀ i, U i ≤ V)
  (⊆ : (V : Set ↑(Spa A A+)) ⊆ ⋃ i, (U i : Set ↑(Spa A A+)))
```

```

{x y : ↑(limitSections V)},
(∀ i, limitRestrict (hle i) x = limitRestrict (hle i) y) → x = y
glue : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
{U : ι → Opens ↑(Spa A A+)}
(hle : ∀ i, U i ≤ V)
(⊆ : (V : Set ↑(Spa A A+)) ⊆ ⋃ i, (U i : Set ↑(Spa A A+)))
(s : ∀ i, ↑(limitSections (U i))),
(∀ i j : ι,
  limitRestrict (inf_le_left (a := U i) (b := U j)) (s i) =
  limitRestrict (inf_le_right (a := U i) (b := U j)) (s j)) →
  ∃ x : ↑(limitSections V), ∀ i, limitRestrict (hle i) x = s i
isEmbedding : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
{U : ι → Opens ↑(Spa A A+)}
(hle : ∀ i, U i ≤ V)
(⊆ : (V : Set ↑(Spa A A+)) ⊆ ⋃ i, (U i : Set ↑(Spa A A+))),
Topology.IsEmbedding (limitRestrictProd hle)

```

The fields `injective` and `glue` are the usual sheaf axioms on arbitrary opens. The final field says that restriction identifies the sections over V homeomorphically with the subspace of $\prod_i \mathcal{O}(U_i)$ consisting of tuples whose restrictions agree. Together these conditions give the concrete form of Wedhorn’s sheaf condition for topological rings.

Remark A.3 (Comparison with Mathlib). For complete Tate pairs the rational-basis and all-open predicates are equivalent:

```

theorem isSheafy_iff_isLimitSheaf :
  IsSheafy A ↔ IsLimitSheaf A := by
  ...

```

The forward implication is the sheaf-on-a-rational-basis theorem, and the converse is restriction to rational covers. For a bundled presheaf F , `F.IsSheafOfTopologicalRings` means that, for every topological commutative ring T , the presheaf $U \mapsto \text{Hom}_{\text{cont}}(T, F(U))$ is a sheaf of sets. This is Wedhorn’s representable formulation [21, Remark 8.20]:

```

def IsSheafOfTopologicalRings : Prop :=
  ∀ (T : Type u) (⊆ : CommRing T) (⊆ : TopologicalSpace T)
  (⊆ : IsTopologicalRing T), ∀ {ι : Type u} (U : ι → Opens X)
  (f : ∀ i, {g : T →+* (F.obj (op (U i))) : Type u} //
    Continuous g),
  (∀ i j,
    (F.map (homOfLE (inf_le_left (a := U i) (b := U j))) (op)).1.comp
      (f i).1 =
      (F.map (homOfLE (inf_le_right (a := U i) (b := U j))) (op)).1.comp
        (f j).1) →
    ∃! g : {g : T →+* (F.obj (op (iSup U))) : Type u} //
      Continuous g,
    ∀ i, (F.map (homOfLE (le_iSup U i)) (op)).1.comp g.1 = (f i).1

```

For the structure presheaf the formalisation proves:

```

theorem structurePresheaf_isSheafOfTopologicalRings_iff :
  TopCat.Presheaf.IsSheafOfTopologicalRings
  (structurePresheaf A) ↔
  IsLimitSheaf A := by
  ...

```

The preceding theorem, together with `isSheafy_iff_isLimitSheaf`, gives

$$\text{IsSheafy } A \iff (\text{structurePresheaf } A).\text{IsSheafOfTopologicalRings}.$$

Thus the bundled `IsSheafOfTopologicalRings` condition and the rational-cover condition used in the proofs are equivalent; neither is merely a consequence of the other.

There is also a direct comparison with Mathlib’s categorical definition. For an arbitrary presheaf F of complete separated topological commutative rings, the general comparison theorem is

```
theorem isSheafOfTopologicalRings_iff_forgetToTopCommRingCat_isSheaf :
  F.IsSheafOfTopologicalRings ↔
    TopCat.Presheaf.IsSheaf
      (F >> CompleteTopCommRingCat.forgetToTopCommRingCat) := by
  ...
```

The functor in this statement forgets completeness and separatedness, but not the topology. The objects of `TopCommRingCat` are all topological commutative rings, so Mathlib’s categorical condition tests exactly the presheaves $U \mapsto \text{Hom}_{\text{cont}}(T, F(U))$ appearing in Wedhorn’s definition. By contrast, the categorical condition with values left in `CompleteTopCommRingCat` tests only complete separated rings and is only a consequence of Wedhorn’s condition.

Specialising the comparison to the structure presheaf gives the all-open equivalence. Its statement is

```
theorem structurePresheaf_forgetToTopCommRingCat_isSheaf_iff :
  TopCat.Presheaf.IsSheaf
    (structurePresheaf A >>
      CompleteTopCommRingCat.forgetToTopCommRingCat) ↔
    IsLimitSheaf A := by
  ...
```

Combining this with the rational-basis theorem above gives the final comparison:

```
theorem isSheafy_iff_structurePresheaf_forgetToTopCommRingCat_isSheaf
  [DecidableEq A] [DecidableEq (RationalLocData A)] [IsTateRing A]
  [IsRingOfIntegralElements (A+ : Subring A)] [T2Space A]
  [NonarchimedeanRing A]
  [let I : UniformSpace A :=
    IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A] :
  IsSheafy A ↔
    TopCat.Presheaf.IsSheaf
      (structurePresheaf A >>
        CompleteTopCommRingCat.forgetToTopCommRingCat) := by
  ...
```

Thus the rational-cover predicate, the all-open predicate, Wedhorn’s Hom-sheaf condition, and Mathlib’s categorical predicate are equivalent. We use `IsSheafy` because it is the convenient form for the arguments in this paper, not because it defines a weaker or stronger notion of sheafiness.

A.3. Choice-independent sheafiness for Tate rings. For a complete Tate ring the following definitions fix a ring of integral elements and then remove the choices of plus ring and completion. They are in `SheafyRing.lean`.

```
def IsSheafyFor (Aplus : RingOfIntegralElements A) : Prop :=
```

```

letI := Aplus.toPlusSubring
haveI : IsRingOfIntegralElements (A+ : Subring A) := Aplus.2
haveI : HasLocLiftPowerBounded A := hasLocLiftPowerBounded_faithful
IsLimitSheaf A

def IsSheafyComplete : Prop :=
  ∀ Aplus : RingOfIntegralElements A, IsSheafyFor A Aplus

def IsSheafyTateRing : Prop :=
  ∀ (P : PairOfDefinition A)
    (Bplus : RingOfIntegralElements (CompletionModel A P)),
    IsSheafyFor (CompletionModel A P) Bplus

```

The predicate `IsSheafyFor A Aplus` says that the Huber pair (A, A^+) is sheafy. For complete Tate rings, the chosen-plus-ring equivalence proves that the choice of integral elements does not affect the condition. The equivalences in `SheafyCompletionModel.lean` remove the choice of completion. Consequently `IsSheafyTateRing` is the analytic specialisation of Wedhorn’s Definition 8.26 [21, Definition 8.26].

A.4. Strong noetherianity.

Definition A.4 (Strong noetherianity). We use the following definitions. They are from `RestrictedPowerSeries.lean`.

```

def MvPowerSeries.IsRestricted {k : ℕ} {A : Type*}
  [CommRing A] [TopologicalSpace A]
  (f : MvPowerSeries (Fin k) A) : Prop :=
  Tendsto (fun s : Fin k →0 ℕ => MvPowerSeries.coeff s f)
    cofinite (nhds 0)

class IsStronglyNoetherian (A : Type*) [CommRing A]
  [TopologicalSpace A] [NonarchimedeanRing A] : Prop where
  isNoetherianRing_restricted : ∀ k : ℕ,
    IsNoetherianRing (restrictedMvPowerSeriesSubring k A)

```

The cofinite convergence condition is the usual condition that the coefficients tend to zero, so `restrictedMvPowerSeriesSubring k A` represents $A\langle T_1, \dots, T_k \rangle$. The second declaration therefore says that these rings are noetherian for every k . Since A is already complete, this is Wedhorn’s definition of strong noetherianity [21, Proposition and Definition 6.36].

A.5. The finite-jet example.

Definition A.5 (The finite-jet ring). We finally record the definition of the example itself. In the following code, `PowerSeries.Restricted R 1` is the radius-one Tate algebra in one variable over R , `RestrictedLaurent K` is $K\langle W, W^{-1} \rangle$, and `DualNumber R` is $R[Q]/(Q^2)$. The source is `FJP/Over/JetRings.lean`.

```

abbrev L : Type u := RestrictedLaurent K

abbrev JetC : Type u := PowerSeries.Restricted (L K) (1 : ℝ)

abbrev JetB : Type u :=
  DualNumber (PowerSeries.Restricted K (1 : ℝ))

abbrev JetD : Type u := DualNumber (L K)

```

```

noncomputable def jetSupport : Subring (JetC K) where
  carrier := {f |
    qCoeff K 0 f ∈ nonnegSubring K ∧
    qCoeff K 1 f ∈ nonnegSubring K}
  zero_mem' := by
    ...
  one_mem' := by
    ...
  add_mem' := by
    ...
  neg_mem' := by
    ...
  mul_mem' := by
    ...

abbrev JetA : Type u := ↑(jetSupport K)

```

The predicate `nonnegSubring K` cuts out $K\langle W \rangle$ inside $K\langle W, W^{-1} \rangle$. Consequently the carrier condition says that the coefficients of Q^0 and Q^1 have no negative powers of W , while the coefficients of Q^n for $n \geq 2$ are unrestricted. Thus $\text{JetA } K$ is

$$\left\{ f_0(W) + Qf_1(W) + Q^2h(W, W^{-1}, Q) \mid \begin{matrix} f_0, f_1 \in K\langle W \rangle \\ h \in L\langle Q \rangle \end{matrix} \right\},$$

which is the coefficient description of A in (3) after inverting ϖ .

The file also constructs the four maps in the Milnor square and proves `milnorRow_exact`: every compatible pair in $\text{JetB } K$ and $\text{JetC } K$ comes from a unique element of $\text{JetA } K$. This identifies $\text{JetA } K$ with $B \times_D C$. The sheaf argument produces the finite-rational-cover condition in the following theorem:

```

include  $\varpi$  hK0 in
theorem isSheafy_JetA :
  ValuationSpectrum.IsSheafy (JetA K) := by
  ...

```

Here ϖ is a `Uniformizer K`, and the remaining input is the noetherianity of `unitBall K`. This is the version of sheafiness proved directly for the example. The equivalences above give the all-open and choice-independent statements. The main theorems are collected in `SheafyEndpoints`, and include the following.

```

theorem finiteJet_isDomain : IsDomain (JetA K) := by
  ...

theorem finiteJet_not_noetherian :
  ¬ IsNoetherianRing (JetA K) := by
  ...

theorem finiteJet_isUniform_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  TopologicalRing.IsUniform (JetA K) := by
  ...

theorem finiteJet_not_stablyUniform_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  ¬ TopologicalRing.IsStablyUniform (JetA K) := by
  ...

```



```

theorem finiteJet_isSheafyFor_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  IsSheafyFor (JetA K) (finiteJetPlus K) := by
...

theorem finiteJet_isSheafyTateRing_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  IsSheafyTateRing (JetA K) := by
...

```

Here K is any complete nontrivially normed ultrametric field for which $\mathcal{O}[K] = \{x : \|x\| \leq 1\}$ is a discrete valuation ring. No local compactness, finiteness of the residue field, characteristic, or perfection assumption is used. The theorem `finiteJet_isSheafyFor_of_dvr` proves that (A, A°) is sheafy, whilst `finiteJet_isSheafyTateRing_of_dvr` removes the choices of completion and ring of integral elements. The corresponding all-open statement is

$$\text{TopCat.Presheaf.IsSheafOfTopologicalRings} \\ (\text{ValuationSpectrum.structurePresheaf (JetA K)}),$$

The corresponding Lean theorem proves this statement directly.

Strong sheafiness is expressed by the following theorem from `FJP/StrongSheafy.lean`.

```

variable (F : Type*) [Field F]

theorem finiteJet_tateExt_isSheafyComplete (n : ℕ) :
  letI := mvTateAlgebraTopology' (A := JetA F) n
  haveI := mvTate_isTateRing (A := JetA F) n
  haveI := mvTate_t2Space (A := JetA F) n
  haveI := mvTate_nonarchimedean (A := JetA F) n
  haveI := mvTateAlgebraTopology'_isTopologicalRing
    (A := JetA F) n
  haveI : @CompleteSpace
    ↑(restrictedMvPowerSeriesSubring n (JetA F))
    (IsTopologicalAddGroup.rightUniformSpace _) :=
    finiteJet_tateExt_completeSpace F n
  IsSheafyComplete
    ↑(restrictedMvPowerSeriesSubring n (JetA F)) := by
...

```

The restricted power-series subring in this statement is $A\langle V_1, \dots, V_n \rangle$, with its canonical Tate-algebra topology, and `IsSheafyComplete` says that it is sheafy for every ring of integral elements. This strong-sheafiness endpoint is currently formalised for the Laurent-series model over $F((t))$. The ordinary sheafiness, uniformity, domain and chart theorems above have the more general DVR-base scope used elsewhere in the appendix.

A.6. Scope of the finite-jet formalisation. The paper's A_0 is represented by the Gauss-norm unit ball. The theorem `isPowerBounded_JetA_iff` identifies this unit ball with the power-bounded elements, giving $A^\circ = A_0$. Completeness and the Tate structure are supplied by the corresponding instances. As in the weighted-parity development, the constant-series map from K is explicit and isometric but is not installed as an `Algebra K` instance.

For the bad chart, `chartEquiv` is an exported ring equivalence. The two continuity theorems make it a homeomorphism, and the canonical-map theorem identifies W and Q . It is not bundled as a `K-AlgEquiv`, which is why Proposition 4.1 is stated for complete topological rings.

The results in section 7 are formalised for any complete ultrametric normed field K equipped with a nonzero element ϖ satisfying $\|\varpi\| < 1$; this is the class `IsFJPBase K`.

```
class IsFJPBase (K : Type u) [NormedField K]
  [IsUltrametricDist K] [CompleteSpace K] where
  pseudoUniformizer : K
  pseudoUniformizer_ne_zero : pseudoUniformizer ≠ 0
  norm_pseudoUniformizer_lt_one :
    ||pseudoUniformizer|| < 1
```

Sheafiness is a separate assertion in this newer interface. The corresponding Lean theorem assumes the additional class `IsFJPNoetherianBase K`, which supplies the noetherian Gauss-unit-ball input used in the sheaf proof. The independent `FiniteJetOver` development used for the main theorem proves sheafiness under the complete discretely valued hypothesis of the paper.

Under the weaker `IsFJPBase` hypothesis, the theorem `finiteJet_problem28` records in one statement that $(W; \varpi)$ is rational, its rational subset and section ring are nonempty and nonzero, multiplication by Q^2 is a continuous strict injection with closed image, and Q^2 restricts to zero. The nonempty point is constructed from the norm valuation pulled back along `ev00`, the map

$$f_0(W) + Qf_1(W) + Q^2h \mapsto f_0(0).$$

The separate declarations saying that multiplication by Q^2 is an isometry and that Q^2 restricts to zero are the inputs used below.

```
variable (F : Type*) [NormedField F] [IsUltrametricDist F]
  [CompleteSpace F] [IsFJPBase F]

theorem finiteJet_problem28 :
  (chartDatum F).IsRational ∧
  (rationalOpen (chartDatum F).T (chartDatum F).s).Nonempty ∧
  Nontrivial (presheafValue (chartDatum F)) ∧
  Function.Injective
    (fun a : JetA F => scottishWitness F * a) ∧
  Continuous (fun a : JetA F => scottishWitness F * a) ∧
  IsStrictMap (fun a : JetA F => scottishWitness F * a) ∧
  IsClosed (Set.range
    (fun a : JetA F => scottishWitness F * a) ∧
    (chartDatum F).canonicalMap (scottishWitness F) = 0 := by
  ...
```

For Problem 24, Lean uses the module structure induced by the canonical map to the completed localisation and proves `finiteJet_not_flat_canonicalMap`. The proof is the one given in Proposition 7.1: Q^2 is a non-zero-divisor of A , so flatness would make its image act injectively on the chart; that image is zero and the chart is nonzero. The ordinary ring localisation `Localization.Away` is always flat. The failure is a property of the completion defining `presheafValue`.

The conclusions of Lemma 5.1 are formalised in the normed setting used for the example: E is one of B, C, D , T_E has its Gauss norm, $T_{E,0}$ is its unit ball, and the finite free terms have their sup norms. Positive-degree exactness is transported from the polynomial sequence by flat base change. Closedness in positive degree comes from exactness and continuity, whilst the degree-zero ideal is closed by the noetherian closed-ideal theorem. The closed-range theorem gives a constant $C_{q,E}$ such that every $y \in \text{im}(d_{q,E})$ has a preimage x with $\|x\| \leq C_{q,E}\|y\|$. This

proves strictness, and multiplying such a preimage by a suitable power of ϖ gives the two denominator inclusions (6) and (7).

The Lean proof of Lemma 5.2 uses these inequalities directly. Its localisation proof obtains the topological embedding from a quotient-norm estimate, and proves openness of $C_\alpha \rightarrow D_\alpha$ from the norm-preserving coefficientwise section. The components of Proposition 5.3 are therefore separate declarations. Finally, the development proves the concrete finite-jet transfer `isSheafy_JetA`: it pushes finite rational covers to B, C, D , glues there, and uses the localised Milnor row and a closed-range theorem.

The Koszul argument, the rational-chart calculation, and the transfer of sheafiness through the concrete Milnor square are contained respectively in the restricted-exactness file and the strictness and closed-image file, `FJP/Over/Chart.lean`, and `FJP/Over/SheafTransfer.lean`.

A.7. The weighted-parity example. The second development uses the same restricted-series infrastructure. Its ambient ring is the countable restricted Tate algebra

$$K\langle W, U_1, U_2, \dots \rangle.$$

The definitions `wpWeight`, `WPMem`, and `wpSupport` are the direct translation of the formula for ω in (10), the definition of S in (11), and the description of $\mathcal{A}$ in (14). The relevant part of the definition is as follows.

```
def wpWeight (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : ℕ :=
  ∑ n ∈ t.support,
    if (t n).mod 2 = 1 ∧ n ≠ 0 then w n else 0

def WPMem (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : Prop :=
  wpWeight w t ≤ t 0

abbrev WPA : Type _ := ↑(wpSupport K w)

def idWeight : ℕ → ℕ := fun n => n

abbrev WPAid : Type _ := WPA K idWeight
```

The coordinate $t(0)$ is the exponent of W ; the other coordinates are the exponents of the U_n . Thus `WPMem idWeight t` says exactly that $a \geq \omega(\nu)$, and `wpSupport` requires the coefficients outside this monoid to vanish. It follows that `WPAid K` is the ring $\mathcal{A}$ of (14). The maps `constA` and `norm_constA` give the isometric embedding of K as the constant series. This map is explicit in the development, although it is not bundled as an `Algebra K` instance.

Suppose that K is complete and nontrivially normed and that $\mathcal{O}[K]$ is a DVR. The principal exported statements are:

```
theorem weightedParity_isUniform_of_dvr :
  IsUniform (WPAid K) := by
  ...

theorem weightedParity_isDomain :
  IsDomain (WPAid K) := by
  ...

theorem weightedParity_not_noetherian :
  ¬ IsNoetherianRing (WPAid K) := by
  ...
```

```

theorem weightedParity_powerBounded_eq_unitBall
  (w : Uniformizer K) :
  powerBoundedSubring (WPAid K) =
    (FiniteJet.unitBall (WPAid K) : Set (WPAid K)) := by
  ...

theorem weightedParity_isSheafyComplete_of_dvr :
  IsSheafyComplete (WPAid K) := by
  ...

theorem weightedParity_chart_isDomain
  (w : Uniformizer K)
  (hK0 : IsNoetherianRing (FiniteJet.unitBall K)) :
  IsDomain (presheafValue
    (chartDatum (w := idWeight) w)) := by
  ...

theorem weightedParity_chart_not_isUniform
  (w : Uniformizer K)
  (hK0 : IsNoetherianRing (FiniteJet.unitBall K)) :
  ¬ IsUniform (presheafValue
    (chartDatum (w := idWeight) w)) := by
  ...

theorem weightedParity_not_stablyUniform_of_dvr :
  ¬ IsStablyUniform (WPAid K) := by
  ...

```

The strong-sheafiness statement is in `WP/StrongSheafy.lean`:

```

theorem wp_tateExt_isSheafyComplete_of_dvr
  [IsDiscreteValuationRing O[K]] (s : ℕ) :
  letI := mvTateAlgebraTopology' (A := WPA K w) s
  haveI := mvTate_isTateRing (A := WPA K w) s
  haveI := mvTate_t2Space (A := WPA K w) s
  haveI := mvTate_nonarchimedean (A := WPA K w) s
  haveI := mvTateAlgebraTopology'_isTopologicalRing
    (A := WPA K w) s
  haveI : @CompleteSpace
    ↑(restrictedMvPowerSeriesSubring s (WPA K w))
    (IsTopologicalAddGroup.rightUniformSpace _) :=
    wp_tateExt_completeSpace (w := w) s
  IsSheafyComplete
    ↑(restrictedMvPowerSeriesSubring s (WPA K w)) := by
  ...

```

Taking $w = \text{idWeight}$, the restricted power-series subring is $\mathcal{A}\langle V_1, \dots, V_s \rangle$. Thus this theorem is precisely the strong sheafiness asserted in Theorem 8.10, for the full DVR-base scope of the paper.

Here `IsSheafyComplete` is the predicate defined above: it quantifies over every ring of integral elements. In particular it proves the sheaf condition for $(\mathcal{A}, \mathcal{A}^\circ)$. The all-open theorem also records the corresponding statement directly for the structure presheaf. The datum `chartDatum` is $(W; \varpi)$, and `chartDatum_isRational` checks that it defines the chart $|W| \leq |\varpi|$. The two chart theorems then say precisely that its section ring is a domain but is not uniform.

The proof routes are the ones used in section 8. Multiplicativity of the countable Gauss norm gives the domain and unit-ball statements. For rational data in $\mathcal{A}^{(N)}$, localisation is proved by taking the completed quotient of $\mathcal{A}^{(N)}$, applying its evaluation map to each coefficient in (17), and constructing the inverse by summing coefficients that tend to zero; it does not invoke the Koszul lemma. For sheafiness, one N is chosen for the base datum and all members of a finite cover. Strong noetherianity of $\mathcal{A}^{(N)}$ glues each coefficient, and the restriction embedding shows that the resulting coefficients tend to zero. The final topological embedding is obtained by applying the nonarchimedean open mapping theorem over K to the closed subspace of compatible tuples. Strong sheafiness is transported across the bicontinuous equivalence between a finite Tate extension and the corresponding weighted-parity algebra with shifted weight. For the bad chart, the quotient formed from $\mathcal{A}^{(0)} = K\langle W \rangle$ is $K\langle X \rangle$, and the coefficientwise ring embeds in formal multivariable power series over this domain. The formalisation also contains, under `WeightedParity.PerturbSetup`, a quantitative norm-unit-ball version of the standard perturbation result cited in Lemma 8.5.

These declarations, including the final failure of stable uniformity, have been checked to use only the usual logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

A.8. Mathematical and code provenance. The formal structure presheaf, rational-cover criterion, and the strongly-noetherian sheaf theorem follow Huber’s construction [11, Theorem 2.2 and Lemmas 2.3–2.6] and Wedhorn’s systematic formulation [21, Proposition and Definition 6.36, Remark 8.20, Definition 8.26, Theorem 8.28(b), Corollary 8.32, and Lemmas 8.33–8.34]. The passage between a fixed plus ring and all plus rings follows the standard rational-refinement argument recorded by Kedlaya [12, Lemma 1.6.8 and Remark 1.6.9]. Bambozzi–Kreminizer formulate the general construction in terms of Koszul complexes and prove strict Koszul regularity [3, Notation 4.3, Definitions 4.5–4.6, and Corollary 4.13]. The formalisation does not invoke this result. Positive-degree exactness is instead proved by flat base change from the polynomial sequence; closedness and the nonarchimedean open mapping theorem [14, Lemma 3.1] give constants bounding the norm of a preimage by a fixed multiple of the norm of its image. Multiplication by a suitable power of ϖ then yields the bounded-denominator criterion used by Buzzard–Verberkmoes [4, Lemma 2].

The development is written in Lean 4 [15] on top of `mathlib` [16]. The formalisation-specific files live in `AINTLIB` [1]. `AINTLIB` includes code adapted from William Coram’s restricted-power-series development [6]; its univariate Gauss-norm layer builds on Fabrizio Barroero’s `mathlib` code [5]. It separately uses Bingyu Xia’s multivariable power-series equivalences [9]. The vendored source headers and commit history retain the respective attributions. The finite-jet and weighted-parity definitions, localisation proofs, and sheaf endpoints discussed in this appendix are the `AINTLIB`-specific part of the development.

REFERENCES

- [1] C. Birkbeck and contributors, *AINTLIB: AdicSpaces formalisation*, Lean 4 source code, finite-jet snapshot `b007a4f3d`, weighted-parity snapshot `090a28921`, sheaf-comparison snapshot `d92f96504`, abstract-base Scottish Book snapshot `01116aca6`, and strong-sheafiness snapshot `870d0eed2`, 2026, GitHub repository.
- [2] O. Ben-Bassat and K. Kreminizer, *Non-Archimedean analytic geometry as relative algebraic geometry*, Ann. Fac. Sci. Toulouse Math. (6) **26** (2017), no. 1, 49–126, DOI.
- [3] F. Bambozzi and K. Kreminizer, *On the sheafiness property of spectra of Banach rings*, J. Lond. Math. Soc. (2) **109** (2024), no. 1, e12855, DOI.
- [4] K. Buzzard and A. Verberkmoes, *Stably uniform affinoids are sheafy*, J. Reine Angew. Math. **740** (2018), 25–39.
- [5] F. Barroero, Lean source code, `mathlib` source file.
- [6] W. Coram, Lean source code, GitHub repository.

- [7] K. Fujiwara and F. Kato, *Foundations of Rigid Geometry I*, EMS Monographs in Mathematics, vol. 7, European Mathematical Society, 2018, DOI.
- [8] D. Hansen and K. S. Kedlaya, *Sheafiness criteria for Huber rings*, preprint, version of 23 April 2025, preprint PDF.
- [9] B. Xia, Lean source code, mathlib pull request #36507.
- [10] R. Huber, *Continuous valuations*, Math. Z. **212** (1993), no. 3, 455–477, DOI.
- [11] R. Huber, *A generalization of formal schemes and rigid analytic varieties*, Math. Z. **217** (1994), 513–551, DOI.
- [12] K. S. Kedlaya, *Sheaves, stacks, and shtukas*, Arizona Winter School notes, 2017, course notes.
- [13] K. S. Kedlaya and R. Liu, *Relative p -adic Hodge theory: Foundations*, Astérisque **371** (2015), arXiv:1301.0792.
- [14] M. Kerz, S. Saito, and G. Tamme, *K -theory of non-archimedean rings II*, Nagoya Math. J. **251** (2023), 669–685.
- [15] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in *Automated Deduction—CADE 28*, Lecture Notes in Comput. Sci., vol. 12699, Springer, 2021, 625–635, DOI.
- [16] The mathlib Community, *The Lean mathematical library*, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, ACM, 2020, 367–381, DOI.
- [17] J. Milnor, *Introduction to Algebraic K -Theory*, Annals of Mathematics Studies, vol. 72, Princeton University Press, Princeton, 1971.
- [18] T. Mihara, *On Tate’s acyclicity and uniformity of Berkovich spectra and adic spectra*, Israel J. Math. **216** (2016), no. 1, 61–105.
- [19] K. S. Kedlaya, *The Nonarchimedean Scottish Book*, Problems 7, 24, and 28, available online.
- [20] The Stacks Project Authors, *The Stacks Project*, project website.
- [21] T. Wedhorn, *Adic spaces*, 2019, arXiv:1910.05934.